

camera
motica

Structure from Motion

Slides are from Svetlana Lazebnik



Драконъ, видимый подъ различными углами зрѣнія
По гравюру на мѣди изъ „Oculus artificialis teleiographicus“ Нолл. 1702 года.

1) first get interest points
SIFT/corners etc

2) track these throughout the
frames

- SIFT descriptors / template
matching /

optical flow

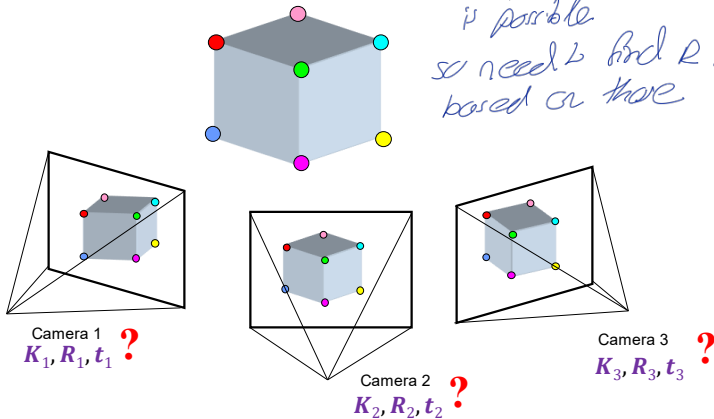
camera $\Rightarrow$ orthographic camera

Outline: Structure from Motion

- Problem definition and ambiguities
- Affine structure from motion
 - Factorization
- Projective structure from motion
 - Bundle adjustment
- Modern structure from motion pipeline

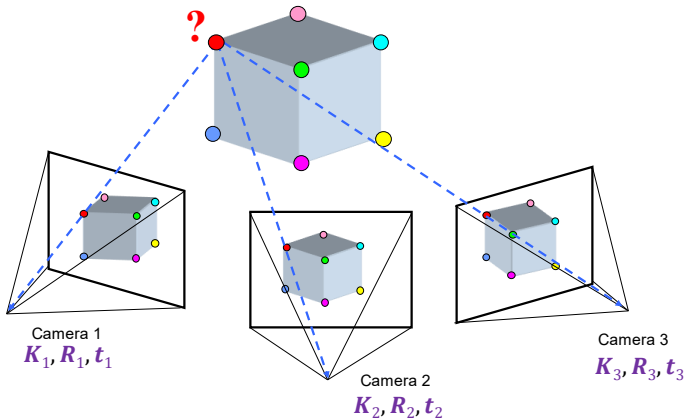
Recall: Calibration

Pinpoint of edges
is possible
so need to find R & t
based on those



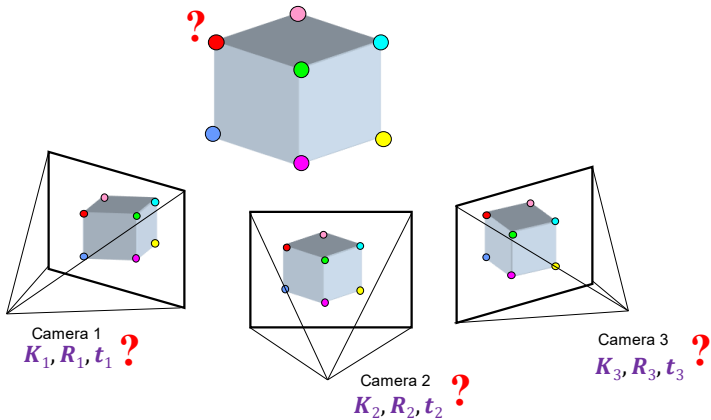
- Given a set of *known* 3D points seen by a camera, compute the camera parameters

Recall: Triangulation



- Given *known cameras* and projections of the same 3D point in two or more images, compute the 3D coordinates of that point

Structure from Motion

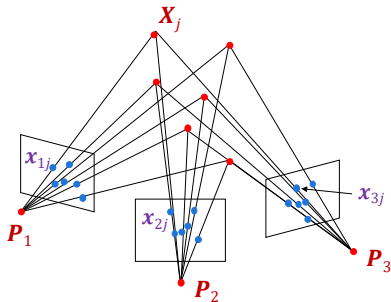


Structure from Motion: Problem Formulation

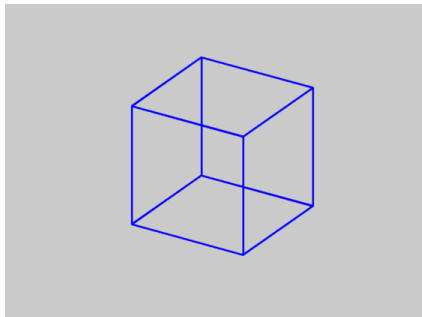
- Given: m images of n fixed 3D points such that (ignoring visibility)

$$\mathbf{x}_{ij} \equiv \mathbf{P}_i \mathbf{X}_j, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

- Problem: estimate m projection matrices $\mathbf{P}_i$ and n 3D points $\mathbf{X}_j$ from the mn correspondences $\mathbf{x}_{ij}$



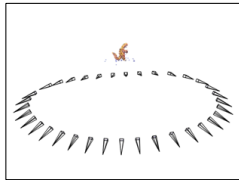
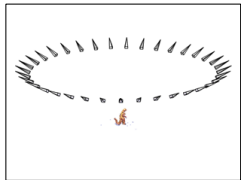
Is SFM always Uniquely Solvable?



Necker cube

Is SFM Always Uniquely Solvable?

- Could actually happen in affine structure from motion:



Source: N. Snavely

Structure from Motion Ambiguity

- If we scale the entire scene by some factor k and, at the same time, scale the camera matrices by the factor of $1/k$, the projections of the scene points remain exactly the same:

$$x \equiv \underset{\substack{\text{cam} \\ \text{matrix}}}{P} \underset{\substack{\text{3D point}}}{X} = \left(\frac{1}{k} P \right) (kX)$$

as can scale both but still get same x

- Without a reference measurement, it is impossible to recover the absolute scale of the scene!
- In general, if we transform the scene using a transformation Q and apply the inverse transformation to the camera matrices, then the image observations do not change:

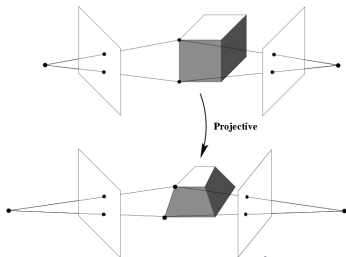
$$x \equiv PX = (PQ^{-1})(QX)$$

Projective Ambiguity

- With no constraints on the camera calibration matrices or on the scene, we can reconstruct up to a *projective* ambiguity:

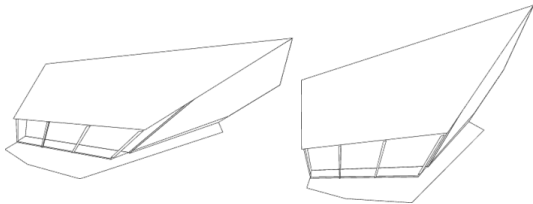
$$x \equiv PX = (PQ^{-1})(QX)$$

Q is a general full-rank 4×4 matrix



same image points
but 3D struct is different

Projective Ambiguity



Affine Ambiguity

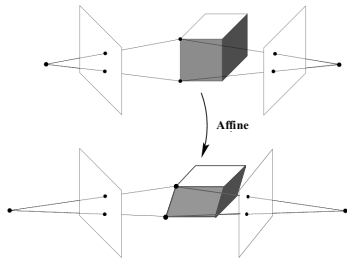
- If we impose parallelism constraints, we can get a reconstruction up to an *affine* ambiguity:

$$x \equiv PX = (PQ_A^{-1})(Q_A X)$$

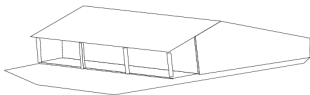
3×3
full-rank
matrix

3×1
translation
vector

$$Q_A = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$$



Affine Ambiguity



parallel lines are parallel but corners are not 90°

Similarity Ambiguity

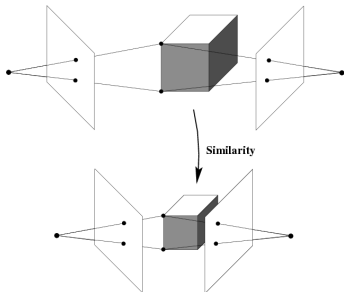
- A reconstruction that obeys orthogonality constraints on camera parameters and/or scene

$$x \equiv PX = (PQ_S^{-1})(Q_S X)$$

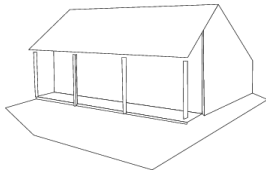
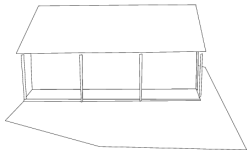
3×3
rotation
matrix

3×1
translation
vector

$$Q_S = \begin{bmatrix} sR & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$$



Similarity Ambiguity

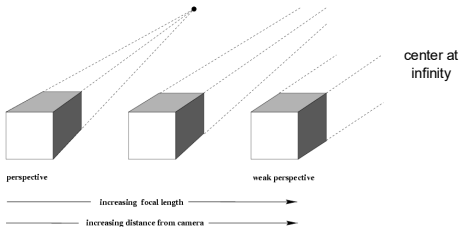
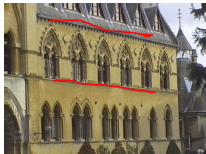


Outline: Structure from Motion

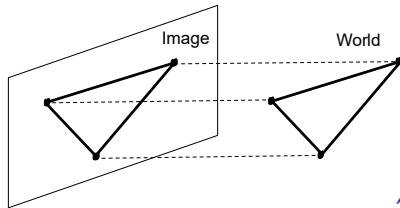
- Problem definition and ambiguities
- Affine structure from motion
 - Factorization

Affine Structure from Motion

- Let's start with *affine* or *weak perspective* cameras



Recall: Orthographic Projection

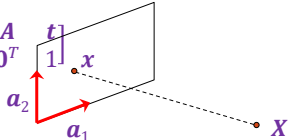


Just drop the z coordinate!

$$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

General Affine Projection

- A general affine projection is a 3D-to-2D linear mapping plus translation:

$$\mathbf{P} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & t_1 \\ a_{21} & a_{22} & a_{23} & t_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$$


- In non-homogeneous coordinates:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} + \begin{pmatrix} t_1 \\ t_2 \end{pmatrix} = \mathbf{A}\mathbf{X} + \mathbf{t}$$

a_1, a_2 : rows of projection matrix

Projection of
world origin

Affine Structure from Motion

- **Given:** m images of n fixed 3D points such that

$$\mathbf{x}_{ij} = \mathbf{A}_i \mathbf{X}_j + \mathbf{t}_i, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

- **Problem:** use the mn correspondences $\mathbf{x}_{ij}$ to estimate m projection matrices $\mathbf{A}_i$ and translation vectors $\mathbf{t}_i$, and n points $\mathbf{X}_j$
- The reconstruction is defined up to an arbitrary *affine* transformation $\mathbf{Q}$ (12 degrees of freedom):

$$\begin{bmatrix} \mathbf{A} & \mathbf{t} \\ 0^T & 1 \end{bmatrix} \rightarrow \begin{bmatrix} \mathbf{A} & \mathbf{t} \\ 0^T & 1 \end{bmatrix} \mathbf{Q}^{-1}, \quad \begin{pmatrix} \mathbf{X}_j \\ 1 \end{pmatrix} \rightarrow \mathbf{Q} \begin{pmatrix} \mathbf{X}_j \\ 1 \end{pmatrix}$$

- How many knowns and unknowns for m images and n points?
 - $2mn$ knowns and $\underbrace{8m}_{\text{cam para.}} + \underbrace{3n}_{\text{2D coord}}$ unknowns
 - To be able to solve this problem, we must have $2mn \geq 8m + 3n - 12$ (affine ambiguity takes away 12 dof)
 - E.g., for **two** views, we need **four** point correspondences

Affine Structure from Motion

- First, center the data by subtracting the centroid of the image points in each view:

$$\begin{aligned}\hat{x}_{ij} &= x_{ij} - \frac{1}{n} \sum_{k=1}^n x_{ik} \\ &= \mathbf{A}_i \mathbf{X}_j + \mathbf{t}_i - \frac{1}{n} \sum_{k=1}^n (\mathbf{A}_i \mathbf{X}_k + \mathbf{t}_i) \\ &= \mathbf{A}_i \left(\mathbf{X}_j - \frac{1}{n} \sum_{k=1}^n \mathbf{X}_k \right) \\ &= \mathbf{A}_i \hat{\mathbf{X}}_j\end{aligned}$$

Affine Structure from Motion

- After centering, each normalized 2D point $\hat{x}_{ij}$ is related to the 3D point by

$$\hat{x}_{ij} = A_i \hat{X}_j$$

- We can get rid of the need to center the 3D data (and the translation ambiguity) by defining the origin of the world coordinate system as the centroid of the 3D points

Affine Structure from Motion

- Let's create a $2m \times n$ data (measurement) matrix:

$$D = \begin{bmatrix} \hat{x}_{11} & \hat{x}_{12} & \cdots & \hat{x}_{1n} \\ \hat{x}_{21} & \hat{x}_{22} & \cdots & \hat{x}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{x}_{m1} & \hat{x}_{m2} & \cdots & \hat{x}_{mn} \end{bmatrix}$$

↓ cameras (2m)

→ points (n)

$$\hat{x}_{ij} = A_i X_j$$

Affine Structure from Motion

- Let's create a $2m \times n$ data (measurement) matrix:

$$D = \begin{bmatrix} \hat{x}_{11} & \hat{x}_{12} & \cdots & \hat{x}_{1n} \\ \hat{x}_{21} & \hat{x}_{22} & \cdots & \hat{x}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{x}_{m1} & \hat{x}_{m2} & \cdots & \hat{x}_{mn} \end{bmatrix} = \begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_m \end{bmatrix} \begin{bmatrix} X_1 & X_2 & \cdots & X_n \end{bmatrix}$$

M cameras $(2m \times 3)$

S points $(3 \times n)$

- What must be the rank of the measurement matrix $D = MS$?

Factorizing the Measurement Matrix

- We want:

The diagram shows the matrix equation $D = M \times S$. Matrix D is a large orange rectangle with a vertical dimension of $2m$ and a horizontal dimension of n . Matrix M is a tall, narrow grey rectangle with a horizontal dimension of 3 . Matrix S is a short, wide red rectangle. The dimensions are indicated by double-headed arrows next to the respective matrices.

$$\begin{matrix} 2m \\ \downarrow \\ \text{orange box } D \\ \uparrow \\ n \end{matrix} = \begin{matrix} \text{grey box } M \\ \downarrow \\ 3 \end{matrix} \times \begin{matrix} \text{red box } S \end{matrix}$$

Factorizing the Measurement Matrix

- Perform SVD of $\mathbf{D}$:

The diagram illustrates the Singular Value Decomposition (SVD) of a matrix $\mathbf{D}$. It consists of four colored rectangular blocks representing matrices, connected by mathematical symbols. From left to right: a dark orange block for $\mathbf{D}$ with dimensions $2m \times n$; an equals sign; a grey block for $\mathbf{U}$ with dimensions $2m \times n$; a multiplication sign; a light orange block for $\mathbf{\Sigma}$ with dimensions $n \times n$; another multiplication sign; and a light orange block for $\mathbf{V}^T$ with dimensions $n \times n$.

$$\mathbf{D}_{2m \times n} = \mathbf{U}_{2m \times n} \times \mathbf{\Sigma}_{n \times n} \times \mathbf{V}^T_{n \times n}$$

Factorizing the Measurement Matrix

Keep top 3 singular values:

$$\begin{array}{c} \text{Orange box} \\ \mathbf{D} \\ 2m \times n \end{array} = \begin{array}{c} \text{Grey box} \quad \text{White box} \\ \mathbf{U}_3 \\ 2m \times 3 \end{array} \times \begin{array}{c} \text{Orange box} \quad \text{White box} \\ \mathbf{\Sigma}_3 \\ 3 \times 3 \end{array} \times \begin{array}{c} \text{Orange box} \quad \text{White box} \\ \mathbf{V}_3^T \\ n \times n \end{array}$$

This is the closest approximation of $\mathbf{D}$ with a rank-3 matrix in terms of Frobenius norm

What to do about $\mathbf{\Sigma}_3$?

One solution: $\mathbf{M} = \mathbf{U}_3 \mathbf{\Sigma}_3^{\frac{1}{2}}, \mathbf{S} = \mathbf{\Sigma}_3^{\frac{1}{2}} \mathbf{V}_3^T$

Factorizing the Measurement Matrix

- One possible solution:

$$\begin{matrix} \text{Orange box} \\ \mathbf{D} \\ 2m \times n \end{matrix} = \begin{matrix} \text{Grey box} \\ \mathbf{M} \\ 2m \times 3 \end{matrix} \times \begin{matrix} \text{Red box} \\ \mathbf{S} \\ 3 \times n \end{matrix}$$

One cpts some ambiguity?

$$\mathbf{S} = \mathbf{\Sigma}_3^{\frac{1}{2}} \mathbf{V}_3^T$$

$$\mathbf{M} = \mathbf{U}_3 \mathbf{\Sigma}_3^{\frac{1}{2}}$$

Are there other solutions?

Factorizing the Measurement Matrix

- Other possible solutions:

$$\begin{matrix} \text{Orange box} \\ \mathbf{D} \\ 2m \times n \end{matrix} = \begin{matrix} \text{Grey box} \\ \mathbf{M} \\ 2m \times 3 \end{matrix} \times \begin{matrix} \text{Pink box} \\ \mathbf{Q} \\ 3 \times 3 \end{matrix} \times \begin{matrix} \text{Pink box} \\ \mathbf{Q}^{-1} \\ 3 \times 3 \end{matrix} \times \begin{matrix} \text{Red box} \\ \mathbf{S} \\ n \times n \end{matrix}$$

We can estimate $\mathbf{Q}$ to give the camera matrices in $\mathbf{M}$ desirable properties, like orthographic projection

Eliminating the Affine Ambiguity

- So far, we have obtained one solution:

$$D = \begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_m \end{bmatrix} \begin{bmatrix} X_1 & X_2 & \cdots & X_n \end{bmatrix}$$

$2n \times 3 \qquad 3 \times m$

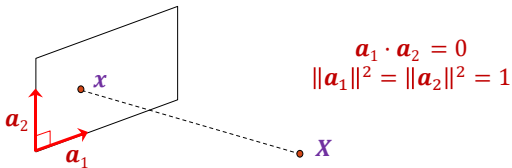
- We want:

$$D = \begin{bmatrix} A_1 Q \\ A_2 Q \\ \vdots \\ A_m Q \end{bmatrix} \begin{bmatrix} Q^{-1} X_1 & Q^{-1} X_2 & \cdots & Q^{-1} X_n \end{bmatrix}$$

such that each camera matrix $A_i Q$ represents orthographic projection, i.e., has orthonormal axes (rows)

Eliminating the Affine Ambiguity

- Let $\mathbf{a}_1$ and $\mathbf{a}_2$ be the rows of a 2×3 orthographic projection matrix. Then



- This translates into $3m$ constraints on the 9 entries of $\mathbf{Q}$:
$$(\mathbf{A}_i \mathbf{Q})(\mathbf{A}_i \mathbf{Q})^T = \mathbf{A}_i (\mathbf{Q} \mathbf{Q}^T) \mathbf{A}_i^T = \mathbf{I}_{2 \times 2}, \quad i = 1, \dots, m$$
 - Are the constraints linear?
 - First, solve for $\mathbf{L} = \mathbf{Q} \mathbf{Q}^T$
 - Recover $\mathbf{Q}$ from $\mathbf{L}$ by Cholesky decomposition
 - Update $\mathbf{M}$ to $\mathbf{M} \mathbf{Q}$, $\mathbf{S}$ to $\mathbf{Q}^{-1} \mathbf{S}$

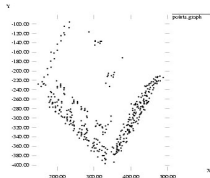
Reconstruction Results



1



60



120



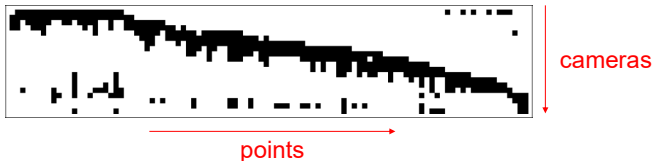
150



C. Tomasi and T. Kanade, [Shape and motion from image streams under orthography: A factorization method](#), IJCV 1992

Dealing with Missing Data

- So far, we have assumed that all points are visible in all views
- In reality, the measurement matrix typically looks something like this:

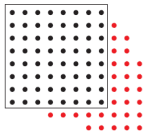


- Possible solution: decompose matrix into dense sub-blocks, factorize each sub-block, and fuse the results
 - Unfortunately, finding dense maximal sub-blocks of the matrix is NP-complete (equivalent to finding maximal cliques in a graph)

Dealing with Missing Data

- Incremental bilinear refinement:

*Skip from
here*



Perform factorization
on a dense sub-block

Solve for a new 3D
point visible by at
least two known
cameras –
triangulation

Solve for a new
camera that sees at
least three known 3D
points – **calibration**

Outline: Structure from Motion

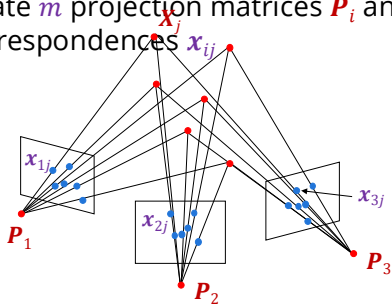
- Problem definition and ambiguities
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- Projective structure from motion

Projective Structure from Motion

- **Given:** m images of n fixed 3D points such that (ignoring visibility):

$$\mathbf{x}_{ij} \equiv \mathbf{P}_i \mathbf{X}_j, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

- **Problem:** estimate m projection matrices $\mathbf{P}_i$ and n 3D points $\mathbf{X}_j$ from the mn correspondences $\mathbf{x}_{ij}$



Projective Structure from Motion

- **Given:** m images of n fixed 3D points such that (ignoring visibility):

$$x_{ij} \equiv P_i X_j, \quad i = 1, \dots, m, \quad j = 1, \dots, n$$

- **Problem:** estimate m projection matrices P_i and n 3D points X_j from the mn correspondences x_{ij}
- With no calibration info, cameras and points can only be recovered up to a 4×4 projective transformation Q :

$$X \rightarrow QX, P \rightarrow PQ^{-1}$$

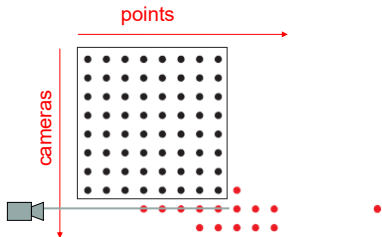
- We can solve for structure and motion when $2mn \geq 11m + 3n - 15$
- For two cameras, at least 7 points are needed

Projective SFM: Two-Camera Case

1. Estimate fundamental matrix F between the two views
2. Set first camera matrix to $[I \mid 0]$
3. Then the second camera matrix is given by $[A \mid t]$ where t is the epipole ($F^T t = 0$) and $A = -[t]_{\times} F$

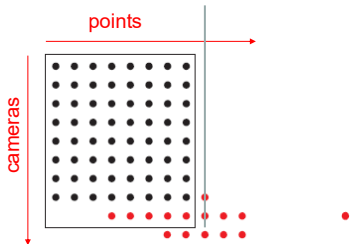
Incremental Structure from Motion

- Initialize motion from two images using fundamental matrix
- Initialize structure by triangulation
- For each additional view:
 - Determine projection matrix of new camera using all the known 3D points that are visible in its image – **calibration**



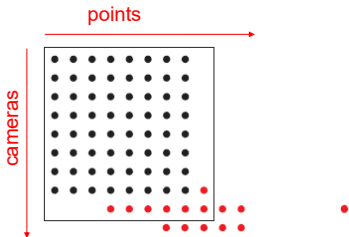
Incremental Structure from Motion

- Initialize motion from two images using fundamental matrix
- Initialize structure by triangulation
- For each additional view:
 - Determine projection matrix of new camera using all the known 3D points that are visible in its image – **calibration**
 - Refine and extend structure: compute new 3D points, re-optimize existing points that are also seen by this camera – **triangulation**



Incremental Structure from Motion

- Initialize motion from two images using fundamental matrix
- Initialize structure by triangulation
- For each additional view:
 - Determine projection matrix of new camera using all the known 3D points that are visible in its image – **calibration**
 - Refine and extend structure: compute new 3D points, re-optimize existing points that are also seen by this camera – **triangulation**
- Refine structure and motion: bundle adjustment



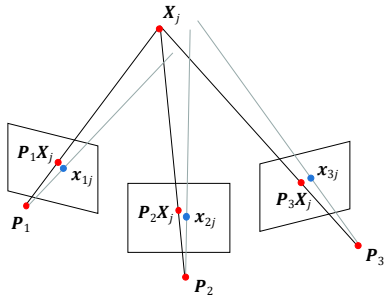
Bundle Adjustment

To here

- Non-linear method for refining structure and motion
- Minimize reprojection error (with lots of bells and whistles):

$$\sum_{i=1}^m \sum_{j=1}^n w_{ij} d(x_{ij} - \text{proj}(\mathbf{P}_i \mathbf{X}_j))^2$$

visibility flag: is
point j visible in
view i ?



Outline: Structure from Motion

- Problem definition and ambiguities
- Affine structure from motion
 - Factorization
- Projective structure from motion
 - Incremental reconstruction, bundle adjustment
- **Modern structure from motion pipeline**

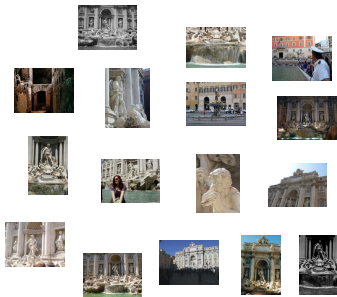
Representative SFM Pipeline



N. Snavely, S. Seitz, and R. Szeliski. [Photo tourism: Exploring photo collections in 3D](http://phototour.cs.washington.edu/). SIGGRAPH 2006
<http://phototour.cs.washington.edu/>

Feature Detection

Detect SIFT features

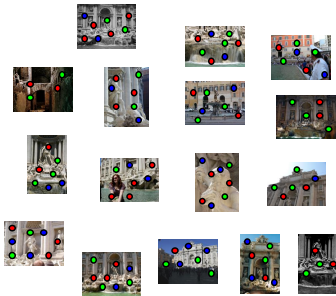


Source: N. Snavely

Feature Detection

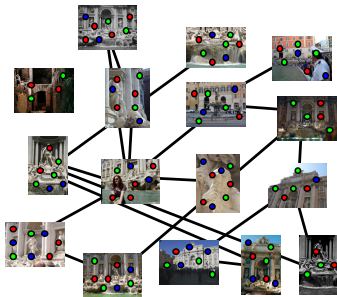
Detect SIFT features

Other popular feature types: SURF, ORB, BRISK, ...



Feature Matching

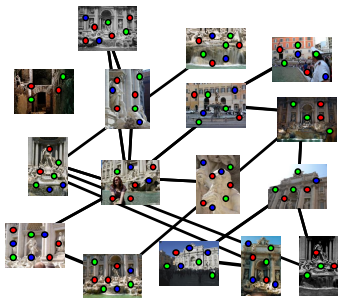
Match features between each pair of images



Source: N. Snavely

Feature Matching

Use RANSAC to estimate fundamental matrix between each pair



Source: N. Snavely

Feature Matching

Use RANSAC to estimate fundamental matrix between each pair



[Image source](#)

Feature matching

Use RANSAC to estimate fundamental matrix between each pair

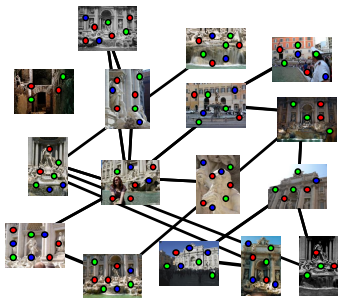
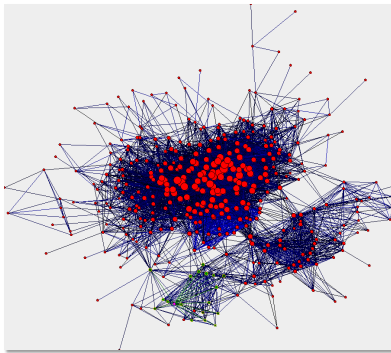


Image Connectivity Graph



(graph layout produced using the Graphviz toolkit: <http://www.graphviz.org/>)

Source: N. Snavely

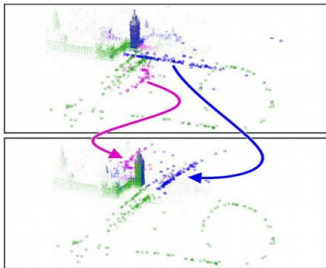
Incremental SFM

- Pick a pair of images with lots of inliers (and preferably, good EXIF data)
 - Initialize intrinsic parameters (focal length, principal point) from EXIF
 - Estimate extrinsic parameters (R and t) using [five-point algorithm](#)
 - Use triangulation to initialize model points
- While remaining images exist
 - Find an image with many feature matches with images in the model
 - Run RANSAC on feature matches to register new image to model
 - Triangulate new points
 - Perform bundle adjustment to re-optimize everything
 - Optionally, align with GPS from EXIF data or ground control points

The Details

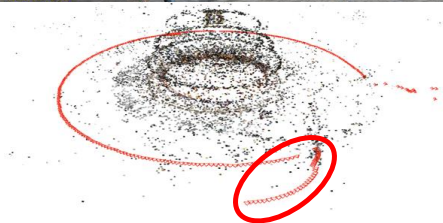
- Handling degenerate configurations (e.g., homographies)
- Filtering out incorrect matches
- Dealing with repetitions and symmetries

Repetitive Structures Cause Catastrophic Failures



<https://demuc.de/tutorials/cvpr2017/sparse-modeling.pdf>

Repetitive Structures Cause Catastrophic Failures



R. Kataria et al. [Improving Structure from Motion with Reliable Resectioning](#). 3DV 2020

Repetitive Structures Cause Catastrophic Failures

Erroneously Matching Images



Baseline Reconstruction



Our Reconstruction



The Details

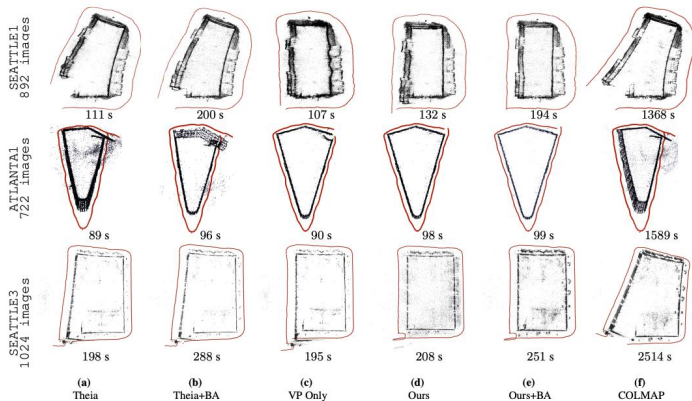
- Handling degenerate configurations (e.g., homographies)
- Filtering out incorrect matches
- Dealing with repetitions and symmetries
- Reducing error accumulation and closing loops

Reducing Error Accumulation and Closing Loops



A. Holynski et al. [Reducing Drift in Structure From Motion Using Extended Features](#). arXiv 2020

Reducing Error Accumulation and Closing Loops



A. Holynski et al. [Reducing Drift in Structure From Motion Using Extended Features](#). arXiv 2020

The Details

- Handling degenerate configurations (e.g., homographies)
- Filtering out incorrect matches
- Dealing with repetitions and symmetries
- Reducing error accumulation and closing loops
- Making the whole thing efficient!
 - See, e.g., [Towards Linear-Time Incremental Structure from Motion](#)

SfM Software

- [Bundler](#)
- [OpenSfM](#)
- [OpenMVG](#)
- [VisualSfM](#)
- [COLMAP](#)
- See also [Wikipedia's list of toolboxes](#)